Adapting to obesity with adipose tissue inflammation

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Abstract | Adipose tissue not only has an important role in the storage of excess nutrients but also senses nutrient status and regulates energy mobilization. An overall positive energy balance is associated with overnutrition and leads to excessive accumulation of fat in adipocytes. These cells respond by initiating an inflammatory response that, although maladaptive in the long run, might initially be a physiological response to the stresses obesity places on adipose tissue. In this Review, we characterize adipose tissue inflammation and review the current knowledge of what triggers obesity-associated inflammation in adipose tissue. We examine the connection between adipose tissue inflammation and the development of insulin resistance and catecholamine resistance and discuss the ensuing state of metabolic inflexibility. Finally, we review the current and potential new anti-inflammatory treatments for obesity-associated metabolic disease.

Biogenic amines
Endogenous molecules with one or more amine groups; five important neurotransmitters are biogenic amines, including three catecholamines (dopamine, noradrenaline and adrenaline) as well as histamine and serotonin.

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The worldwide epidemic of obesity has led to a dramatic increase in the metabolic diseases associated with this condition, which has focused a great deal of attention on the underlying mechanisms of obesity and its co-morbidities. Evidence suggests that many of the co-morbidities of obesity, including type 2 diabetes mellitus, nonalcoholic fatty liver disease, steatohepatitis, asthma, cancer, cardiovascular diseases and neurodegenerative diseases, are related to the generation of low-grade, chronic inflammation[1–3]. The trigger for this inflammation is uncertain, and the causal relationship between inflammation and the complications of obesity remains somewhat in question; however, little doubt remains that obesity is closely associated with inflammation and that the degree of inflammation correlates well with the severity of insulin resistance and type 2 diabetes mellitus[1–4], which suggests that understanding the inflammatory response might lead to the development of new approaches for treating these devastating diseases.

Chronic inflammation often presents in three stages[4] (FIG. 1). An initial trigger, usually a stressor of some kind, is followed by an acute, adaptive inflammatory response and then a long-term maladaptive phase that leads to complications. In the case of obesity, the initial trigger might result from homeostatic stress produced by a positive energy balance and an overall hyper-anabolic state, particularly in adipocytes. These cells respond by releasing chemokines that initiate an adaptive inflammatory response, enabling healthy expansion of adipocytes while simultaneously reducing energy storage, all of which occurs at the expense of homeostasis. However, over time, the system strives to restore homeostasis, which can be accomplished only by achieving a new set point for weight, blood levels of glucose, sympathetic tone, circulating levels of lipids and hormone levels. These changes are accompanied by reduced metabolic flexibility, long-term insulin and catecholamine resistance, abnormal tissue remodelling and fibrosis.

Adipocytes have primary roles in controlling energy homeostasis. These cells are not only the first choice for energy storage and utilization, but also sense energy needs, and secrete hormones and lipids that coordinate to regulate other tissues. One such hormone is leptin, which is secreted from adipocytes when a high energy state is reached and works centrally to reduce food intake and increase energy expenditure[5–11]. Here, we review the current knowledge of adipose tissue inflammation and focus on the transition from an adaptive to a maladaptive state. We discuss the important role of adipocytes in energy homeostasis and how adipose tissue responds to overnutrition, detailing the changes in innate immunity that lead to some of the deleterious effects of sustained obesity. We review the current understanding of the cellular and molecular triggers for obesity-induced inflammation and discuss potential therapeutic approaches for addressing this problem.

Sensing and managing energy status

Although energy balance is controlled by numerous hormones and biogenic amines, adipose tissue has an important role as both a storage depot and a sensor of energy storage status. Once consumed, nutrients induce secretion of insulin, which instructs adipocytes and myocytes to transport and store those nutrients as triglycerides

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Key points

- Adipocytes have an important role in sensing and managing energy status
- Adipose tissue responds to overnutrition by mounting an immune response; however, the initial inflammatory trigger in adipose tissue is unknown
- Inflammation induces insulin resistance through a variety of molecular mechanisms
- The maladaptive responses that occur in long-term obesity are a result of chronic inflammation, particularly catecholamine resistance
- Inflammatory pathways are intriguing therapeutic targets for metabolic disease; however, the clinical efficacy of drugs targeting these pathways has been disappointing

or glycogen[12]. Adipocytes are particularly important in this process, as they also sense energy storage in a cell-autonomous manner and in turn secrete hormones, such as leptin, to initiate a feedback loop that reduces food intake and activates the sympathetic nervous system[5–9]. This latter action results in the release of adrenaline and noradrenaline from nerve terminals in adipose tissue, activating β-adrenergic receptors on adipocytes, which increases the rate of lipolysis and thermogenic processes[5,6,10]. This closed-loop brain–adipocyte axis is one example of how adipocytes control both the storage and the mobilization of energy via crosstalk with other tissues[11]. Other adipocyte-secreted factors, including adiponectin and fibroblast growth factor 21, as well as cytokines and inflammatory proteins, can also have important effects on systemic energy metabolism[13–21].

In addition to functioning in energy storage and sensing, adipocytes can also directly contribute to energy expenditure via thermogenesis. Brown adipocytes, which are characterized by a high mitochondrial content and multilocular lipid droplets, are professional thermogenic cells. Brown adipose tissue is abundant in rodents and infant humans, and some evidence indicates that variable amounts of this type of adipose tissue are present in adult humans[22–25]. The identification of beige adipocytes in rodents and humans has introduced more complexity to the thermogenic program[22,24,26]. Despite having a lineage similar to that of white adipocytes, beige adipocytes exhibit many of the thermogenic characteristics of brown adipocytes[24,27,28]. Visceral adipose tissue remains fairly stable as a white adipose depot and is a primary site at which lipid storage and mobilization occur via lipogenesis and lipolysis, respectively. By contrast, adipocytes with beige characteristics are likely to develop within subcutaneous white adipose depots[29–31]. Differentiation of beige adipocytes could be controlled by various afferent and hormonal signals in response to energy status[32], and the innate immune system might also be involved in this process[33].

Response to overnutrition

During conditions of a sustained positive energy balance, the closed-loop brain–adipocyte axis described above can be pushed to its limits, which results in hypersecretion of insulin, leptin, catecholamines and other hormones. Adipocytes expand in size and number to accommodate the need for increased lipid storage and the anabolic force of hyperinsulinaemia. However, in most cases, these cells ultimately reach a threshold at which further anabolic pressure cannot be accommodated owing to constraints on cell and tissue expansion. Reaching this threshold creates stress on adipocytes, and one response to this stress is to initiate an inflammatory program.

Mesenteric adipose tissue
A small membrane-like visceral fat depot connected to the intestine that expands into a large adipose depot surrounding the intestine during obesity.

Triggers of the inflammatory program

Although the precise triggers of obesity-associated inflammation are poorly understood, several potential mechanisms have emerged (FIG. 2). The trigger for adipose tissue inflammation could emanate from a gut-derived substance, dietary component or metabolite. Alternatively, the rapid expansion of adipose tissue in obesity could provide intrinsic signals that might trigger an inflammatory response; these signals include adipocyte death, hypoxia and mechanotransduction arising from interactions between the cell and the extracellular matrix (ECM).

Gut-derived antigens. Obesity is thought to give rise to increased intestinal permeability, which results in higher circulating levels of lipopolysaccharide produced by intestinal Gram-negative bacterial species[34]. This gut-derived lipopolysaccharide might initiate an inflammatory cascade via the activation of pattern recognition receptors (PRRs), such as Toll-like receptor 4 (TLR4), in adipocytes[35]. Gut-derived lipopolysaccharide could be an important inflammatory trigger, particularly in visceral fat, or might amplify the effects of an earlier inflammatory trigger. Inflammation in mesenteric adipose tissue, which surrounds the gut, is essential for its expansion in response to overnutrition[36,37]. In turn, expansion of this adipose depot might be a protective factor that prevents the spread of gut-derived antigens systemically, as mice with defects in mesenteric expansion (which develop when mice are given a high-fat diet) exhibit more inflammation and worse metabolic disease than wild-type mice[37]. Specifically, overexpression of the anti-inflammatory adenoviral protein complex RIDα/β in adipocytes results in failure to accumulate mesenteric adipose tissue in response to a high-fat diet[37]. In addition to increased insulin resistance, these mice also exhibit more gut permeability and increased serum levels of anti-lipopolysaccharide immunoglobulin Gs[37]. The presence of these antibodies indicates leakiness of the gut, which enables lipopolysaccharide to enter the blood stream, and an inflammatory response to this antigen. Consistent with this theory, higher circulating levels of lipopolysaccharide are observed in humans with type 2 diabetes mellitus than in controls[38].

Dietary or endogenous lipids. Various lipid species that are elevated owing to diet or obesity might also contribute to adipose tissue inflammation. Free fatty acids promote inflammation by binding to TLRs, such as TLR4 and TLR2 (REFS 39,40), which promote downstream NF-κB signalling[35]. Once activated, NF-κB can increase the synthesis and secretion of chemokines, such as MCP1 (also known as CCL2), by adipocytes, which leads to infiltration of pro-inflammatory macrophages (see below). The expression of both TLR2 and

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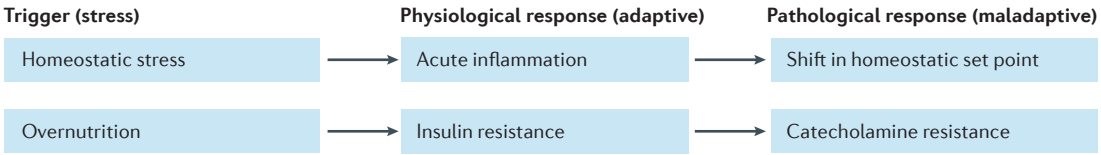

Figure 1 | **Adaptive and maladaptive phases of inflammation in metabolic disease.** In chronic inflammatory diseases, an initial stress gives rise to a physiological adaptive response intended to alleviate the stress. Over time, the adaptive response morphs into a maladaptive response that has pathological effects. In the case of obesity, the initial trigger is homeostatic stress triggered by the anabolic pressure of a positive energy balance. The adaptive inflammatory response is catabolic, and alleviates the anabolic pressure and supports the expansion of adipose tissue. However, over time, the system strives to re-establish homeostasis with a new set point for weight, blood levels of sugar and other hormones, culminating in the pathological effects of long-term obesity.

TLR4 is increased in the adipose tissue of individuals with obesity, indicating that these receptors are involved in obesity-associated inflammatory signalling[41,42]. Other potential sensors of free fatty acids include lipid chaperones, such as FABP4, which might promote pro-inflammatory signalling in macrophages[43]. Some lipid species, such as omega-3 essential fatty acids, have anti-inflammatory roles, and evidence suggests that the actions or levels of these lipids might be reduced by certain diets or modified indirectly by obesity[44–47]. The role of omega-6 essential fatty acids in inflammation is unclear, as these lipid species can be used in synthetic pathways to produce pro-inflammatory moieties[46,48]. Although not essential fatty acids, dietary omega-9 fatty acids also seem to be beneficial and are associated with reduced inflammation[49,50]. An omega-9 fatty acid, oleic acid, is responsible for many of the anti-inflammatory effects of olive oil and is an important factor in the health benefits observed with the Mediterranean diet[51,52].

Adipocyte death. Dead adipocytes, which accumulate in the adipose tissue of those with obesity, are readily apparent by histology, as they appear in the centres of crown-like structures containing F4/80-positive immune cells[53]. Dead and dying adipocytes send out numerous signals that contribute to the development of inflammation in adipose tissue[54,55]. Pro-inflammatory macrophages are recruited from blood monocytes to adipose tissue[56], whereas resident classically activated macrophages proliferate in crown-like structures and then move throughout adipose tissue[57]. These processes contribute to the dramatic increase in the number of macrophages observed in the adipose tissue of people with obesity (discussed below). The NOD-like receptor (NLR) family of PRRs also sense obesity-induced signals, such as damage-associated molecular patterns, that originate from stressed adipocytes[58]. NLRs are activated by stressed or dying cells and mobilize leukocytes towards these stimuli to limit tissue damage[59,60]. In macrophages, activation of NLRs activates the NALP3 inflammasome (also known as cryopyrin, encoded by *NLRP3*) to induce the production of IL-1β and IL-18 via caspase 1 (REF. 61). Diet-induced obesity also induces the production of caspase 1 and IL-1β in adipose tissue, and *Nlrp3*-deficient and *Casp1*-deficient mice are resistant to obesity-induced inflammation[61,62].

Hypoxia. Substantial evidence suggests that hypoxia develops as adipose tissue expands and that the reduction in oxygen tension as a result of decreased perfusion of adipose tissue or increased metabolism might be an initiator of inflammation[63]. Obesity in rodents and patients is well known to induce localized reductions in PO_2 (REF. 64), which is associated with increased expression of *Glut1*, *Leptin*, *Il6*, *Vegf* and *Pai1* (REFS 65–68). Moreover, exposure of adipose tissue in culture to hypoxic conditions can induce changes in gene expression, including the upregulation of numerous genes that are associated with inflammation[69]. Increased levels of hypoxia-inducible factor 1α (encoded by *Hif1a*) have been observed in adipose tissue from obese rodents, and immunostaining has revealed that areas of hypoxia are correlated with regions displaying infiltration of macrophages[70]. While the molecular events involved in the initiation or actions of adipose tissue hypoxia have not yet been elucidated, evidence indicates that signalling through the NF-κB pathway might be increased in hypoxic adipose tissue[71]. However, despite substantial evidence of a link between hypoxia and inflammation, it remains uncertain whether hypoxia is simply a consequence of adipose tissue expansion or a direct causative contributor to obesity-associated metabolic disease.

Mechanical stress. Another potential mechanism for induction of inflammation is mechanical stress on the adipocyte. Adipocytes interact with their basement membrane and ECM via pathways that govern differentiation and expansion in response to obesity. Cell shape directly and potently regulates gene expression in numerous cell systems via relays that depend on the actin cytoskeleton[72]. Adipocytes are embedded in a dense network of ECM proteins, particularly collagen 1, which is highly crosslinked in adipose tissue[73]. Increased storage of triglycerides in adipocytes in the ECM-fixed environment can elicit various mechanical stresses on these cells. Indeed, knockout of genes that encode collagens[74] or the collagenases that degrade collagen, particularly MMP14, has a major effect on adipocyte function, lipid synthesis and storage, and overall energy metabolism[75], particularly *in vivo* or in a 3D setting *in vitro*[75]. Although the pathways that are controlled by mechanical stress in adipocytes have not been elucidated, RhoA, ROCK and NF-κB signalling pathways have been evaluated.

Crown-like structures
These histological structures are observed in adipose tissue and are the result of immune cells surrounding dead adipocytes; as the immune cells pack into the spaces around the dead adipocytes and between the surrounding live adipocytes, they form a shape reminiscent of a crown.

Oxygen tension
The partial pressure of oxygen within a tissue, which is a measure of the amount of oxygen within the tissue.

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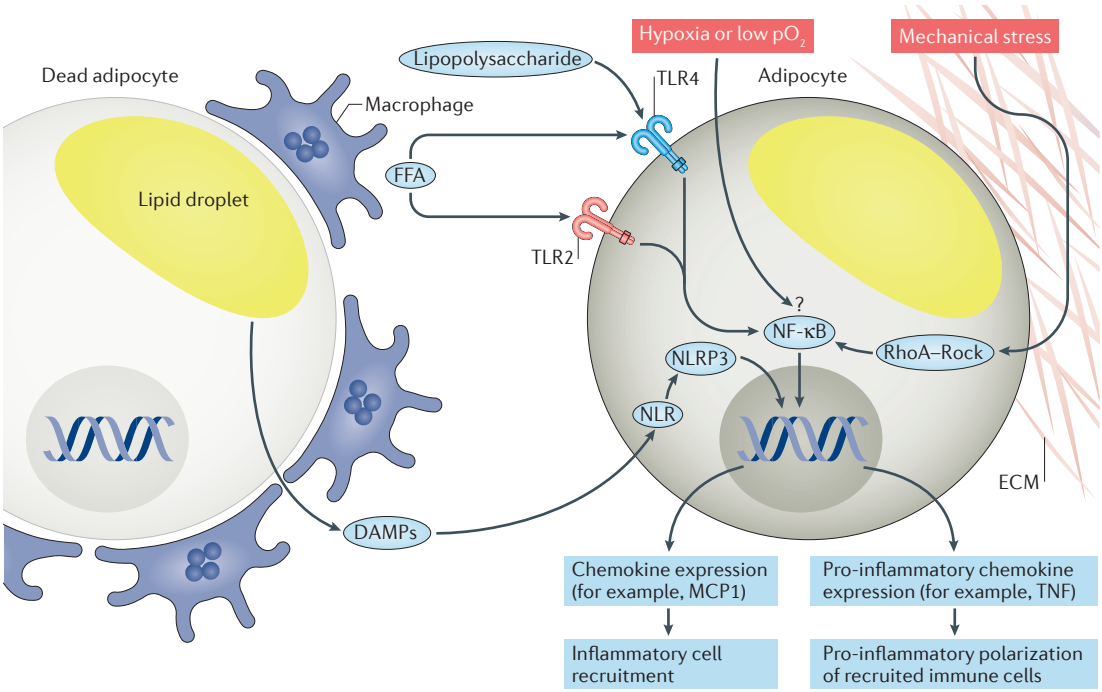

Figure 2 | **Initiators of obesity-associated inflammation in adipocytes.** Gut-derived lipopolysaccharide might stimulate inflammatory pathways by binding to the pattern recognition receptor Toll-like receptor 4 (TLR4) at the plasma membrane. Similarly, free fatty acids (FFAs) can activate inflammatory signalling through either TLR4 or TLR2. Additionally, NLRP3 activation by damage-associated molecular proteins (DAMPs), which are released from dying adipocytes and recognized by NOD-like receptors (NLRs), might also be an important initiating step in inflammation. Hypoxic conditions are also associated with inflammation in adipocytes, but the exact mechanisms responsible for this association are unclear. Finally, mechanical stress caused by adipose tissue expansion through the extracellular matrix (ECM) is sensed by the RhoA–Rock pathway, which leads to downstream inflammatory signalling. NF-κB is a signalling hub that has been suggested to be involved in inflammatory signalling downstream of all these diverse potential initiators of adipocyte inflammation in individuals with obesity. Expression of genes that encode downstream inflammatory proteins leads to expression of adipokines, including inflammatory cytokines that promote the recruitment and activation of pro-inflammatory immune cells. TNF, tumour necrosis factor.

Adipogenesis
The differentiation of pre-adipocytes (or other fibroblast-like cells) to make new adipocytes.

Type 2 or T helper 2 (T_H2)
Type 2 cytokines, such as IL-4 and IL-13, are classified based on their secretion from T_H2 cells (however, they are secreted from many other immune cells) and are typically anti-inflammatory and promote M2 macrophage differentiation; type 2 immunity is characterized by antibody-mediated immune responses.

Type 1 or T helper 1 (T_H1)
Type 1 cytokines, such as IL-1β and tumour necrosis factor, are classified based on their secretion from T_H1 cells (however, they are secreted from many other immune cells) and are typically pro-inflammatory; type 1 immunity is characterized by a phagocytic immune response.

RhoA signalling, along with cell shape, seems to be a negative regulator of both adipogenesis and adipocyte hypertrophy, whereas ECM density affects inflammation via NF-κB signalling[76–78].

The immune response in adipose tissue

All three types of adipose depots (white, brown and beige) usually contain immune cells that surveil and maintain the integrity and hormonal sensitivity of adipocytes. In lean animals, these immune cells operate in an overall Type 2 or T helper 2 (T_H2) state, in which master regulators, such as IL-33, which is secreted primarily from epithelial cells, engage in a coordinated effort to control tissue integrity and metabolism. IL-33 induces innate lymphoid cells (ILC2s) to generate cytokines, such as IL-5 and IL-13, which in turn can activate eosinophils[79,80]. Eosinophils are found in adipose tissue and secrete IL-4, which maintains macrophages in an M2-polarized (or alternately activated) state[81] and induces the differentiation of beige adipocytes[82,83]. Numerous studies have shown that M2 macrophages depend on PPARγ and PPARδ, which activate M2 pathways that induce the expression of genes that encode anti-inflammatory proteins, such as arginase[84,85]. Moreover, the cytokine IL-10 (which is produced by M2 macrophages) preserves the insulin sensitivity of adipocytes and in turn suppresses lipolytic signals, thus ensuring that the system retains its metabolic flexibility[86]. Thus, under normal energy balance conditions, adipocytes and immune cells coordinate to regulate the storage or mobilization of energy in response to the organism's needs.

In contrast to lean animals, the immune cells in the adipose tissue of obese animals operate in an overall Type 1 or T helper 1 (T_H1) state, in which master regulators, such as tumour necrosis factor (TNF), which is secreted from adipocytes or immune cells, engage in a coordinated effort to preserve tissue integrity while adapting to the metabolic needs associated with overnutrition. Numerous studies indicate that macrophages undergo dramatic changes in the context of obesity. For instance, obesity is associated with an overall increase in the number of macrophages in both rodents and humans[87], largely owing to the recruitment of M1-polarized (or pro-inflammatory) macrophages, which adopt a pro-inflammatory phenotype[56] and secrete cytokines such as TNF and IL-1β. The combination of an increase in total macrophages and an increased ratio of M1 to M2 macrophages is a hallmark of the adipose tissue

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inflammation that accompanies obesity and is associated with the development of insulin resistance and metabolic diseases[56,88].

Increased inflammation in the adipose tissue of individuals with obesity is the result of a higher proportion of pro-inflammatory macrophages in the tissue rather than an increase in inflammation within individual macrophages[89]. The pro-inflammatory macrophages that appear in large numbers in the adipose tissue of people with obesity are often referred to as M1 macrophages owing to their pro-inflammatory phenotype; however, just as obesity-associated chronic low-grade inflammation differs from inflammation associated with acute infection, macrophage inflammation in response to obesity is not identical to the classic M1 activation state. As macrophages in adipose tissue express a distinct set of surface markers, the pro-inflammatory activation that occurs in obesity has been referred to as metabolic activation, or Me, rather than M1 (REF. 90). Not surprisingly, the pathways required for chronic low-grade inflammation are distinct from the pathways involved in the inflammatory response to infection. The pro-inflammatory macrophages observed in obese adipose tissue can also be distinguished from classically activated macrophages by their higher expression of genes that encode proteins involved in lipid metabolism[89]. For example, in the obese state, both M1 and M2 macrophages in adipose tissue have a gene expression profile that is associated with liposomal lipid metabolism[89]. However, the lysosomal program of gene expression does not seem to drive the inflammatory response to obesity[89]. In addition, crosstalk occurs between the pathways that control metabolism and inflammation. For example, the lipid sensitive nuclear receptor PPARγ, which is active in macrophages from obese adipose tissue, might limit inflammation and maintain adipose tissue in a state of chronic low-grade inflammation[84,85,91].

During the early stages of obesity, macrophages serve as effectors of a complex immune program. In addition to $Cd4^+$ T_H1 cells and T_H2 cells, regulatory $Cd8^+$ T effector cells, which produce IFNγ, also seem to have an important role in obesity-induced adipose tissue inflammation. Obesity is associated with an increase in the number of $Cd8^+$ T effector cells in mice, but the relevance of these cells in humans is unclear; although some studies have found increased numbers of $Cd8^+$ T cells in the adipose tissues of humans with obesity, others have not[92–95]. Importantly, the infiltration of T effector cells was found to precede the accumulation of macrophages in adipose tissue[93]. Furthermore, inhibition of $Cd8^+$ T effector cells with an anti-Cd8 antibody resulted in reduced M1 macrophage numbers in the adipose tissue of obese mice[93]. Obesity is also associated with an increase in the number of $Cd4^+$ T_H1 cells relative to the number of T_H2 cells and regulatory T (T_{reg}) cells[92,96]. Reductions in the number of T_{reg} cells during obesity are thought to contribute to the development of adipose tissue inflammation[97]; however, evidence also suggests that the increase in T_{reg} cells that occurs during ageing contributes to ageing-associated inflammation and insulin resistance[98].

B cell accumulation has been observed to precede T cell accumulation during the development of obesity[99]. This observation raises the intriguing possibility that antigen presentation by B cells might contribute to the development of obesity-associated inflammation. B cells can be genetically ablated by knockout of the immunoglobulin μ heavy-chain[100], and B cell-null mice exhibit reduced adipose tissue inflammation in the context of diet-induced obesity[101]. MHC-I positive B cells contribute to IFNγ production in $Cd8^+$ T cells, and MHC-II-positive B cells contribute to IFNγ production in $Cd4^+$ T cells. Consistent with an antigen presentation mechanism of action, direct cell-to-cell contact is required for B cells to promote pro-inflammatory T cell development in a co-culture of the two cell types isolated from patients with type 2 diabetes mellitus[102]. In addition to regulating T cells, B cells might also promote adipose tissue inflammation through the production of immunoglobulin G antibodies or pro-inflammatory cytokines[101,102]. Similar to T cells and macrophages, different classes of B cells exist. While B2 cells have a pro-inflammatory effect in adipose tissue, B1α cells constitutively express the anti-inflammatory cytokine IL-10 and prevent the development of adipose tissue inflammation in obesity[103]. B cells might not be the only antigen-presenting cells that affect T cell development in obesity. Primary adipocytes from individuals with obesity exhibit an activated program of MHC II antigen presentation[104]. Furthermore, MHC II antigen presentation in adipocytes is an early event in obesity, observed after only 2 weeks of high-fat feeding in mice[104]. Adipocytes might also present antigens that activate invariant natural killer T cells via CD1, which attenuates obesity-induced inflammation[105].

Other tissues, including the liver, also become inflamed during the obese state[106–108]. Inflammation in these other tissues can also contribute to the development of insulin resistance and metabolic diseases. However, adipose tissue inflammation has a unique role in the pathology of obesity. Upon weight loss, inflammation in the liver is resolved; however, adipose tissue seems to have an obesogenic memory and retains its inflammatory state despite weight loss[106]. In humans, as is often the case, this response is highly variable. After extreme weight loss, such as in response to bariatric surgery, some individuals experience resolution of adipose tissue inflammation, whereas others do not[106]. Continued adipose tissue inflammation was associated with adipocyte insulin resistance despite weight loss[106]. The effect that this obesogenic memory in adipose tissue has on obesity-associated morbidity and mortality is unknown; however, it could be related to the observation that being overweight or obese during one's lifetime, regardless of subsequent weight loss, increases mortality[109].

Inflammation-induced insulin resistance

Insulin resistance is defined as the failure to mount a normal physiological response to insulin. This effect manifests mainly as reduced non-oxidative glucose disposal in response to insulin, as well as reduced suppression of lipolysis and hepatic glucose production. Although insulin resistance can lead to type 2 diabetes mellitus, it might

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Box 1 | **Gestational insulin resistance**

Insulin resistance in adipose tissue is probably a counter-regulatory mechanism intended to promote nutrient mobilization and reduce adiposity. This phenomenon is readily apparent during late pregnancy, when the development of insulin resistance reduces glucose uptake and promotes lipolysis in maternal adipose tissue so that these nutrients can be utilized by the growing fetus. Gestational insulin resistance is considered an essential physiological event, whereas obesity-associated insulin resistance is viewed as pathological; however, they have a common underlying inflammatory mechanism. In fact, the increase in tumour necrosis factor levels that occurs in late pregnancy is an important predictor of the change in insulin sensitivity that occurs at this time[174–176]; no correlations with insulin sensitivity were observed for any of the other placental hormones, including cortisol[174].

initially occur as a physiological response to obesity, which resists the anabolic pressures of insulin intended to reduce excessive nutrient storage. Pregnancy provides a precedent for this physiological role of insulin resistance (BOX 1). Overwhelming evidence now suggests that inflammation is directly associated with insulin resistance[1–4]. As a general observation, individuals with obesity who are insulin resistant exhibit a high degree of adipose tissue inflammation, whereas patients with obesity who remain insulin sensitive do not exhibit adipose tissue inflammation[110,111]. Numerous molecular signalling pathways have been described at the interface between inflammation and metabolism. Inhibition of inflammatory signalling by knockout of key pathways in obese mice, including components of the NF-κB[112] and c-Jun N-terminal kinase (JNK)[113] pathways, as well as inhibition or knockout of numerous other pro-inflammatory signalling molecules, scaffolding proteins and cytokines, can disrupt the link between obesity and insulin resistance[61,114–121]. Many of these pathways can directly or indirectly block insulin action. The key players in promoting insulin resistance are likely to be cytokines such as TNF and IL-1β, which are locally secreted from pro-inflammatory adipose tissue macrophages[1]. Activation of inflammatory pathways can also prompt the release of other factors from macrophages that might attenuate insulin action, including LTB4 (REF. 122) and galectin[123].

As mentioned above, the ability of PRRs to sense endogenous ligands whose expression is induced under obesogenic conditions is thought to have a key role in obesity-associated inflammation. For instance, TLR4-deficient mice are protected from the activation of the inflammatory response that is induced by obesity, as well as from the insulin resistance induced by lipid infusion[115]. Furthermore, TLR2-knockout mice are also protected from the insulin resistance associated with diet-induced obesity, which suggests that TLRs (nearly all members of this family are expressed in adipose tissue) might have a broad role in the development of obesity[124]. The downstream effects of PRR activation might also include regulation of intracellular lipid species, such as ceramides and sphingolipids[125]. Inhibition of ceramide production blocks the ability of saturated fatty acids to induce insulin resistance[114]. Although the mechanisms underlying ceramide-induced insulin resistance remain uncertain, blockade of the activation or increased dephosphorylation of Akt could have a key role[126–128].

Many of these obesity-generated inflammatory signals converge to activate serine kinases that directly block insulin action. Among these is JNK, which is activated in response to stress signals, such as TLRs, fatty acids and inflammatory cytokines[113,129–132]. Some controversy surrounds the precise tissues in which JNK is activated and affects metabolism in response to obesity; however, adipocytes do seem to be involved in this process[129]. Most data suggest that JNK, and perhaps other stress-activated kinases, have a role in blocking insulin receptor signalling by serine/threonine phosphorylation of insulin receptor substrates, thereby reducing tyrosine phosphorylation and activation of downstream signals[133]. Furthermore, endoplasmic reticulum (ER) stress and downstream activation of the molecular pathways governing the unfolded protein response seem to be closely tied to both JNK1 and IKK/NF-κB pathway activation[116]. Widespread activation of ER stress signalling components and cascades (including ATF6, PERK and IRE1) is seen in obesity, and pharmacologic inhibition of ER stress can reverse metabolic dysfunction[108]. IFN-induced, double-stranded RNA-activated protein kinase (PKR; also known as EIF2AK2) acts as a PRR at the interface between ER stress and nutrient sensing that translates these signals into an inflammatory response through JNK[116].

Maladaptive responses to inflammation

Although the overall degree of inflammation correlates well with the severity of metabolic disease, inflammation might also be required for adaptive responses to overnutrition, as it promotes angiogenesis to prevent hypoxia and induces insulin resistance to limit the rate of energy accumulation in cells[134–137]. Moreover, the expansion of adipose tissue, which requires a low level of inflammatory signalling, is essential for preventing ectopic lipid deposition in other tissues and cells, such as the liver, muscle and pancreatic β cells, where it has toxic effects[36,134,138,139]. These responses to overnutrition involve a complex network of crosstalk among the various cells types within adipose tissue. In addition to adipocytes, adipose tissue contains many other types of cells that regulate dynamic nutrient storage, including preadipocytes, endothelial cells, macrophages and other immune cells. Preadipocytes are primed and ready to differentiate into mature adipocytes to increase the storage capacity of adipose tissue and are also capable of releasing energy-mobilizing hormones, such as IL-6, atrial natriuretic peptide and preadipocyte factor 1, in response to appropriate signals[135,136]. Importantly, adipogenesis is tightly linked with angiogenesis to create space and support the nutrient needs of newly formed adipocytes. Endothelial cells enable the exchange of nutrients between the blood (or lymph) and adipose tissue, mediating the delivery of nutrients and oxygen to support nutrient storage in adipocytes, as well as the mobilization and distribution of lipids released from adipocytes during lipolysis. Additionally, newly recruited immune cells must also cross the endothelial barrier to enter adipose tissue. Thus, this network of cell types in adipose tissue coordinates to control the response to energy needs.

Angiogenesis
The proliferation and migration of endothelial cells to form new blood vessels.

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Metabolic inflexibility
A state in which responsiveness to catabolic and anabolic signals is reduced in adipocytes, leaving them unable to efficiently contract (mobilize nutrients) or expand (store nutrients).

As previously discussed, the initial inflammatory response to increased adiposity (which reduces insulin sensitivity) comes at the expense of maintaining energy homeostasis while also supporting adipose tissue expansion. However, the continued angiogenic and catabolic response to obesity ultimately has a detrimental effect on metabolism. As the organism seeks to maintain blood levels of glucose and energy balance within a narrow range, it must readjust its set point to achieve homeostasis. Thus, despite the marked reduction in sensitivity to the major anabolic hormone, insulin, the organism strives to preserve energy storage by reducing energy expenditure[137]. One way in which this aim is accomplished in adipose tissue is through the development of resistance to lipolytic or thermogenic signals, such as catecholamines or leptin. At the same time, continued unresolved inflammation, angiogenesis and adipose tissue expansion ultimately lead to fibrosis, which is associated with metabolic inflexibility, deregulation of metabolic pathways and adipocyte death.

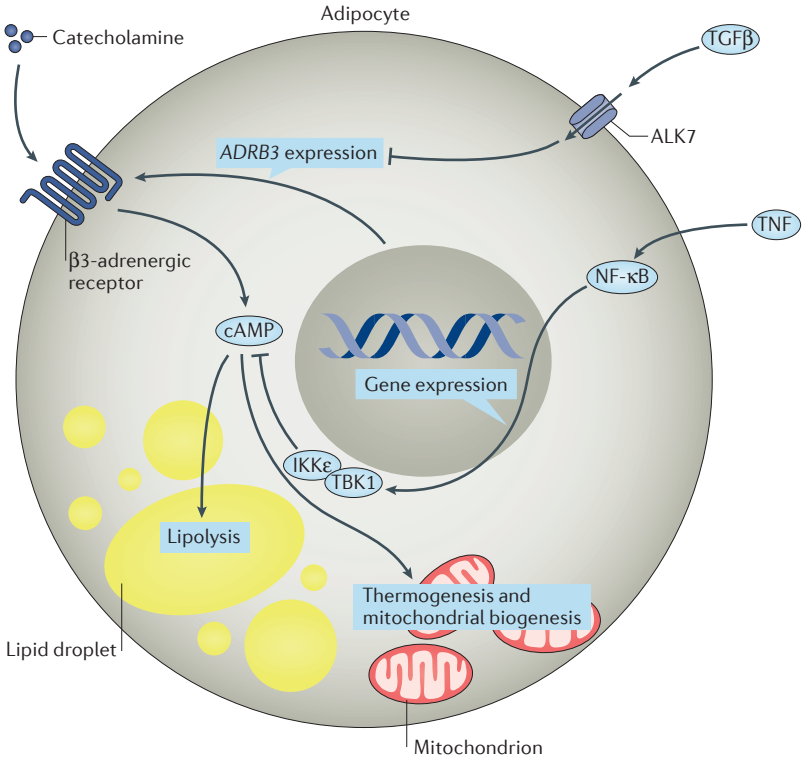

Figure 3 | **Mechanisms underlying obesity-associated catecholamine resistance.** Catecholamines activate β-adrenergic receptors on adipocytes to increase intracellular levels of cAMP, which causes a kinase cascade resulting in lipolysis and the activation of transcription factors, including ATF2 and ATF4, which promote the expression of genes involved in thermogenesis as well as mitochondrial biogenesis. Obesity-associated catecholamine resistance occurs as a result of both reduced expression of the cell surface receptor *ADRB3* and inhibition of intracellular signalling. One mechanism that reduces *ADRB3* expression involves activation of ALK7, a receptor for TGFβ. Another mechanism of catecholamine resistance reduces post-receptor signalling after long-term tumour necrosis factor (TNF) exposure, which increases the expression of the non-canonical IKKs via NF-κB. IKKε and TBK1 then dampen cAMP signalling through the activation of phosphodiesterase 3B.

Catecholamine resistance

Adipocytes are normally regulated by anabolic (insulin) and catabolic (leptin or catecholamine) signals, as discussed above. Several studies have revealed that long-term obesity is associated with resistance to both leptin and catecholamines[140–147]. Failure of adipocytes to respond to adrenergic stimulation enables the preservation of energy storage in the context of insulin resistance. However, resistance to both insulin and catecholamines leaves adipocytes metabolically inflexible[137,140].

Although the underlying causes of leptin resistance remain a mystery, several hypotheses for this phenomenon have been proposed, including downregulation of leptin receptors owing to high concentrations of leptin[141], impairment of the access of leptin to key receptors in the central nervous system[142], reductions in signalling from the leptin receptor or defects in the downstream neural circuitry that responds to the hormone[9,143]. Several studies have also linked hypothalamic inflammation and ER stress to leptin resistance[144,145].

In contrast to the well-studied role of leptin resistance in obesity, the role of catecholamine resistance is often overlooked. Similar to insulin resistance, catecholamine resistance is defined as the failure to mount a normal physiological response to catecholamines. At the molecular level, catecholamine resistance manifests as reduced β-adrenergic signalling, which results in a reduction in lipolysis and thermogenesis in response to sympathetic activation in adipose tissue. In mice, catecholamine resistance has been shown to contribute to the development of obesity[146,147]. Catecholamine resistance is known to occur in humans with obesity and probably contributes to the development of obesity[148–150]. Resistance to the lipolytic effects of catecholamines has been observed in men, women and children with obesity, as well as in *ex vivo* adipocytes[148–150]. Loss of the catabolic response to catecholamines might be a causal factor in the development of human obesity, as was shown in mice; however, this phenomenon has not been examined in prospective human studies.

The mechanisms by which adipocytes become insensitive to stimulation by the hormones that bind to β-adrenergic receptors (catecholamines) in obesity remain uncertain, but several pathways have been implicated, namely, pathways characterized by reduced expression of β2 and β3-adrenergic receptors[151], reduced mitochondrial biogenesis[152], increased expression of the TGFβ receptor ALK7 (REF. 153) and reduced activity of post-receptor pathways, for example, by activation of cAMP phosphodiesterases[154] (FIG. 3). The findings from a paper published in 2016 suggest that some of these effects might be the result of long-term inflammation[152]. While the cytokine–JNK–NF-κB pathways are thought to be catabolic in nature and associated with increased lipolysis, studies have indicated that their long-term activation results in reduced expression of several genes involved in lipolysis and thermogenesis, such as *HSL* (which encodes hormone-sensitive lipase; also known as *LIPE*) and *UCP1* (REF. 154). For example, acute treatment of adipocytes with TNF promotes lipolysis, whereas long-term treatment with TNF causes catecholamine resistance and therefore reduced lipolysis[154]. Expression of the non-canonical

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IκB kinases IKKε and TBK1 is induced as a result of NF-κB activation by cytokines such as TNF[131] and might be associated with the catecholamine resistance observed in obesity. This association could result from the phosphorylation and activation of cAMP phosphodiesterase 3B, which directly reduces the sensitivity of adipocytes to β-adrenergic action[154]. An inhibitor of IKKε and TBK1, amlexanox, restored catecholamine sensitivity[154] and reversed the effects of high-fat feeding in mice, including weight gain, fatty liver and insulin resistance[155]. Thus, inflammation might be linked to reductions in energy expenditure that are associated with obesity through a process that is maladaptive.

Inflammation as a therapeutic target

The correlations among obesity, adipose tissue inflammation and metabolic disease make inflammatory pathways an attractive target for the treatment of metabolic diseases. However, the results of clinical trials of anti-inflammatory agents have thus far been disappointing. For instance, antibodies that neutralize TNF have proven to be an effective treatment for many other inflammatory diseases, such as rheumatoid arthritis[156], and some patients with this disease have experienced beneficial changes in glucose metabolism[157,158]. However, prospective studies of the antibodies that neutralize TNF have been discouraging in patients with type 2 diabetes mellitus[159]. Despite having beneficial effects in mice, the human anti-TNF antibody CDP571 elicited no improvements in insulin sensitivity in a clinical trial involving patients with type 2 diabetes mellitus[160]. Administration of IL-1β antagonists has a beneficial effect on glucose handling in patients with type 2 diabetes mellitus[161]. However, adipose tissue does not seem to be an important target for these agents, as the beneficial effects of the drugs seem to be mediated by the pancreas[161]. Perhaps the greatest progress has been made with salicylate[162]. Independent studies have shown that anti-inflammatory doses of salicylate cause reductions in fasting blood levels of sugar and HbA_{1c}; however, this treatment has no effect on insulin sensitivity[162]. The anti-inflammatory effects of salicylates are thought to be mediated by inhibition of IKKβ; however, salicylates also activate AMP-activated kinase and heat shock proteins[163]. While salicylate treatment does reduce inflammation in the adipose tissue of individuals with obesity, these anti-inflammatory effects do not correlate with metabolic benefits[164]. Salicylate also reduces insulin clearance and might therefore partly improve hyperglycaemia via inflammation-independent mechanisms[165].

Evidence also suggests that type 2 immunity can contribute to thermogenic pathways in brown and beige adipocytes[82,83]. The obese state is characterized by a decrease in levels of a cytokine associated with type 2 immunity, IL-33, which controls the activity and expansion of ILC2s, as well as the anti-inflammatory T cell subtype T_{reg} cells[82,83]. ILC2s can initiate a cascade of events that might ultimately lead to the development of beige adipocytes in white adipose depots[82,83], whereas T_{reg} cells might control pathways that contribute to insulin sensitivity[166]. Thus, activation of type 2 immune signalling could have numerous beneficial effects on metabolism. Determining whether the findings of these rodent studies can be translated to humans will be crucial. However, it should be noted that immune pathways are probably not the primary regulators of energy expenditure in adipose tissue, as this process is governed mainly by catecholamines released from nerve terminals that are centrally controlled.

Similarly, inhibitors of TBK1 and IKKε could be used to treat obesity-dependent metabolic disease. As described above, one such compound, amlexanox, can restore the catecholamine sensitivity of adipocytes both *in vivo* and *in vitro*[154,155]. Alone or in combination with β-adrenergic agonists, amlexanox has the potential to increase energy expenditure and perhaps induce the browning of white adipose tissue, which could improve glucose metabolism[154,155,167]. This compound is currently being evaluated in clinical studies. Preliminary results indicate that amlexanox has beneficial effects on glucose metabolism[168]. Interestingly, these effects are limited to the subset of patients who exhibit higher baseline inflammation in adipose tissue[168].

It should be noted that several well-established drugs used for the treatment of type 2 diabetes mellitus have anti-inflammatory properties. For instance, metformin directly inhibits the production of reactive oxygen species from complex 1 of the mitochondrial electron transport chain and can limit the production of numerous cytokines[169]. Additionally, metformin can block NF-κB signalling via the action of AMP-activated kinase[170]. Thiazolidinediones have dramatic anti-inflammatory effects in macrophages and adipocytes by activating PPARγ[171]. These drugs can block NF-κB[172] and restore the M2 phenotype of macrophages[173].

The reasons for the modest effects of anti-inflammatory therapies observed thus far are not known. One potential problem associated with treating inflammation is that targeting one cytokine, chemokine or pathway might not be sufficient to have the desired effect. Furthermore, a potential issue complicating the interpretation of these data is the lack of biomarker evidence demonstrating whether or not the drugs elicit anti-inflammatory effects as expected. It will be important in future clinical studies to obtain adipose tissue biopsy samples that can be evaluated for changes in protein and gene expression.

Conclusions

The current obesity epidemic has necessitated focusing a great deal of attention on the molecular mechanisms associated with its complications. Investigations conducted over the past decade have revealed a great deal about how cells and tissues respond to the stress of overnutrition and about the interplay between adipose tissue and other cell types that are critically involved in energy homeostasis. These findings suggest that the inflammatory response to obesity might have both beneficial and harmful effects, depending on the stage and degree of obesity, as well as other factors. Thus, it will be important to dissect the crucial molecular events that connect obesity with defects in hormone responsiveness and both energy storage and utilization. Ultimately, these insights might yield interesting, new therapeutic targets with which to block the cycle of obesity and its devastating complications.

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